TERM 1

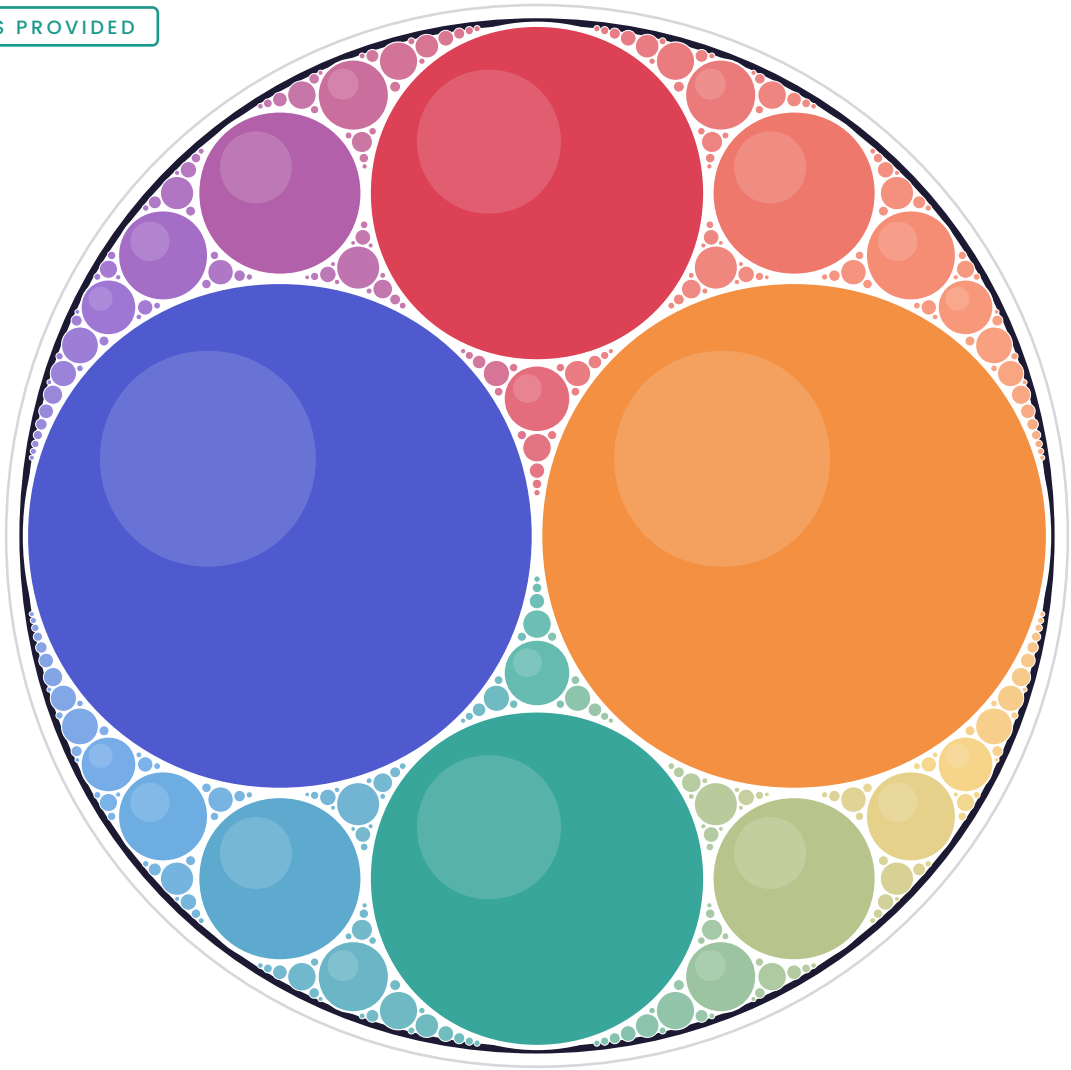
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

APOLLONIAN GASKET
every circle touches three others; every curvature is a whole number

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

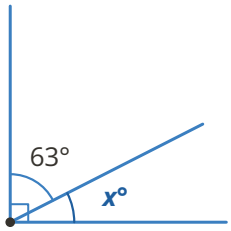
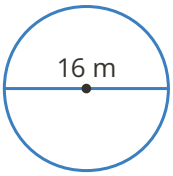
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Evaluate. $15 - 3 \times (2 + 4)$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Expand. $4(m + 9)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mean of these heights. cm: 150, 175, 160, 165, 170	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

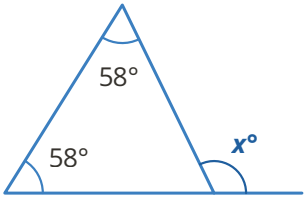
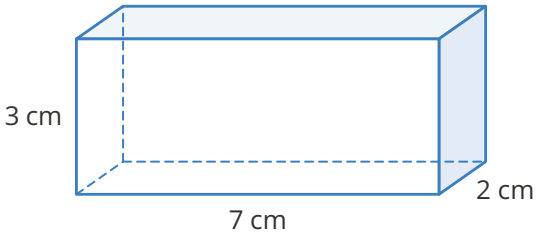
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Calculate. 0.64×10	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Share \$500 in the ratio 3 : 2. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the probability that the surf is not flat. $P(\text{flat}) = \frac{3}{8}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

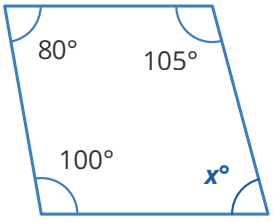
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this rectangle.  	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 12 of the 16 nippers at the surf club passed the swim test. What percentage passed?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Solve. $3x + 5 = 20$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 2x - 2$. x: −1, 0, 1, 2 y: _____, −2, 0, _____	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

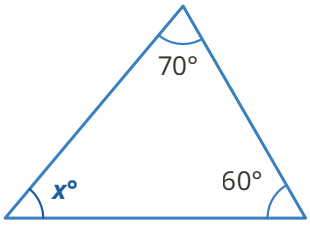
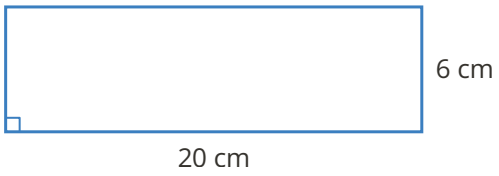
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the perimeter of this rectangle.  	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
3 $\frac{2}{3}$ of the 24 players in a touch footy squad are in Year 8. How many players is that?	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $x = 5$. $x^2 - 2x$	FORMULA $x^2 - 2x$ COPY SUBSTITUTE ANSWER
5 Find the range of these netball scores. Goals: 26, 18, 31, 22, 24	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

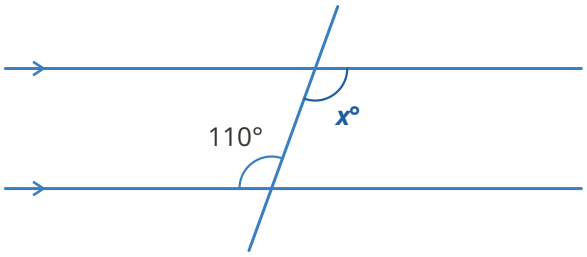
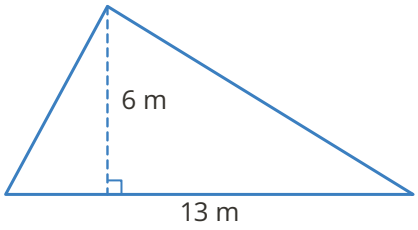
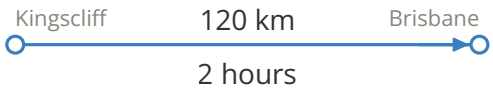
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Alternate angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 25% of 120 L	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 A car travels from Kingscliff to Brisbane. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A letter is chosen at random from this word. Find $P(A)$. B A N A N A	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

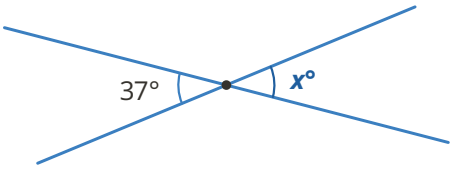
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Convert 4000 m to kilometres.	FORMULA 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 64%	FORMULA Percentage to fraction: write over 100, then simplify COPY SUBSTITUTE ANSWER
4 3 kg of bananas cost \$6. Find the price per kilogram.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 A spinner was spun 100 times. Find the experimental probability of green. Colour: Red | Blue | Green Frequency: 23 | 41 | 36	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

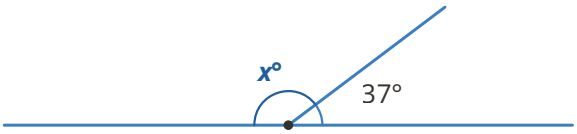
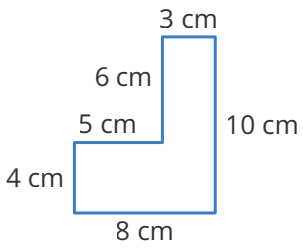
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt[3]{125} \times \sqrt{4}$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Expand. $7(2 + k)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mode of these scores. 14 11 12 14 15 11 14	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

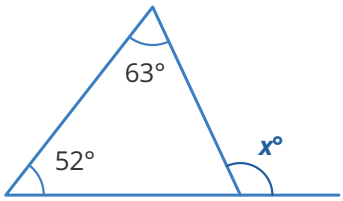
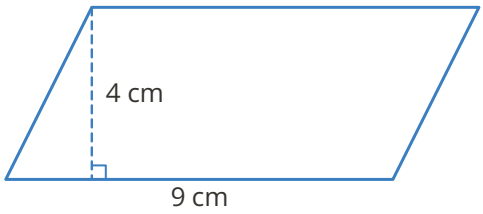
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 19 out of 20	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 A touch footy carnival runs from 10:40 to 12:15. How long is that?	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 Find the mean of these netball scores. Points: 12, 15, 9, 18, 6	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

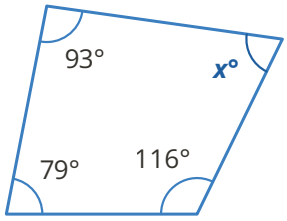
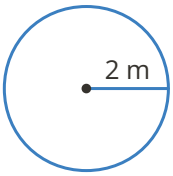
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Evaluate. $-20 \div 5 \times 3$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Solve. $x + 7 = 15$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the probability that it does not rain. $P(\text{rain}) = \frac{2}{5}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

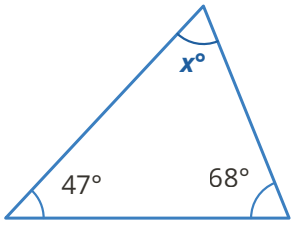
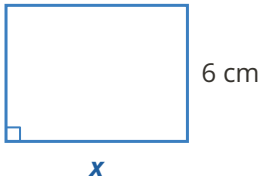
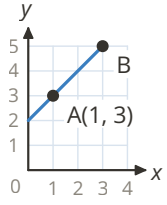
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 The area of this rectangle is 48 cm². Find x. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{5}{6}$ of \$42	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when m = 6 and n = 7. $5m - 2n$	FORMULA $5m - 2n$ COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = x + 2$. Write the coordinates of B. 	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

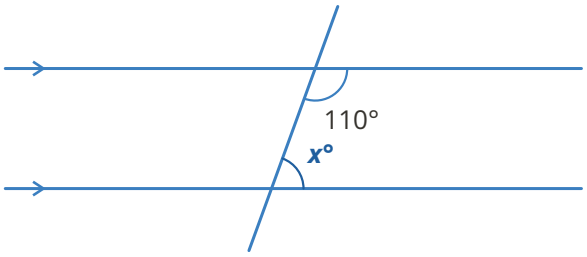
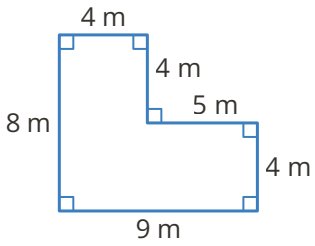
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this shape.  	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Calculate. 75% of $48	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Simplify. 15 : 25	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is green. Colour: Red, Blue, Green Marbles: 4, 2, 6	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

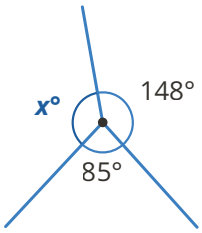
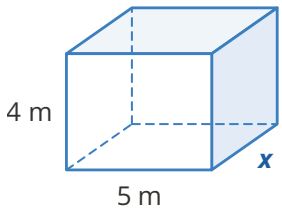
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 The volume of this rectangular prism is 100 m^3. Find x. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. $\frac{9}{10}$	FORMULA Fraction to percentage: fraction × 100% COPY SUBSTITUTE ANSWER
4 Share 84 lollies in the ratio 4 : 3. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the range of these try tallies. Tries: 8, 15, 11, 6, 13	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

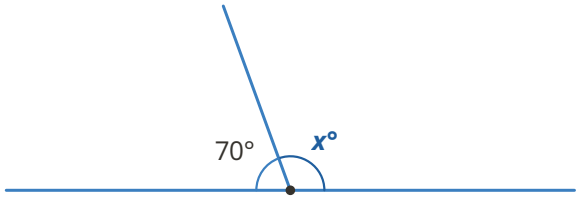
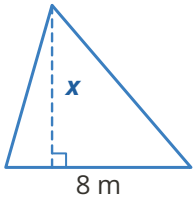
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 The area of this triangle is 28 m². Find x.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $3060 \div 100$	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Expand. $5(3x - 2)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Heads came up 42 times in 100 tosses of a coin. Find the experimental probability of heads.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

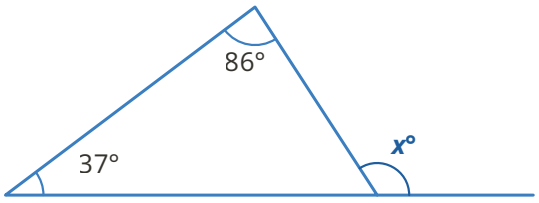
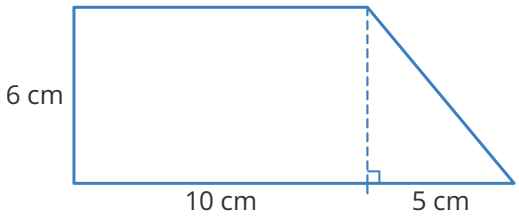
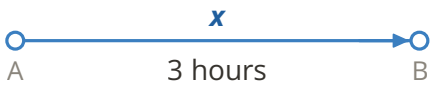
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Write as a percentage. 13 out of 50	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 A car travels from A to B at an average speed of 80 km/h. Find x. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the mean of these goal tallies. Goals: 7, 9, 4, 10, 5	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

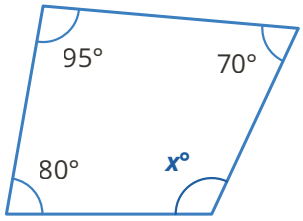
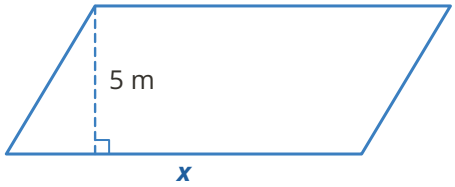
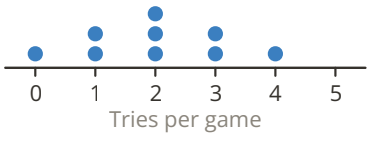
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 The area of this parallelogram is 60 m². Find x. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{7}{8}$ of \$64	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Which bag of rice is the better buy? Bag A: 2 kg for \$6 Bag B: 5 kg for \$20	FORMULA Price per unit = price ÷ number of units; best buy = lowest price per unit COPY SUBSTITUTE ANSWER
5 The dot plot shows the tries a rugby league team scored in each game. Find the median. 	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

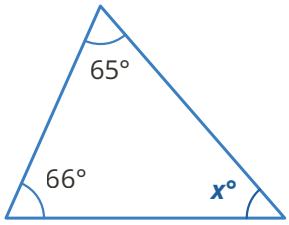
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Convert 120 000 cm^2 to square metres.	FORMULA $1\ m^2 = 10\ 000\ cm^2$ COPY SUBSTITUTE ANSWER
3 Calculate. 50% of 86 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Solve. $\frac{x}{2} - 3 = 5$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the probability that the team does not win. $P(\text{win}) = \frac{7}{10}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

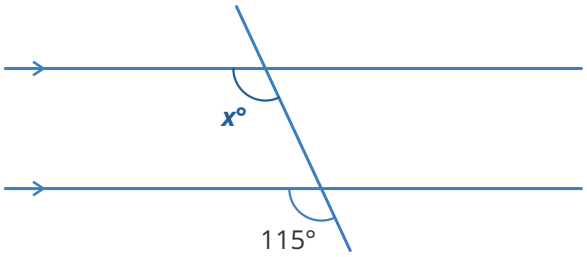
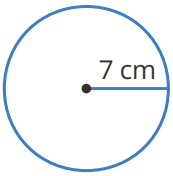
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Evaluate. $9 - (-3) \times 2$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $m = 4$. $3(2m - 1)$	FORMULA $3(2m - 1)$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is black. Colour: Black, White Marbles: 7, 3	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

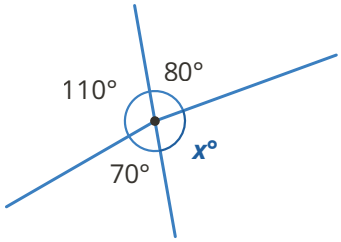
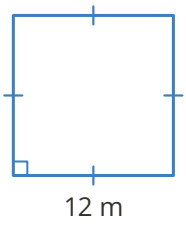
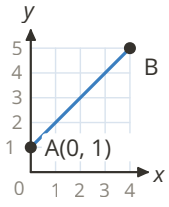
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this square.  	FORMULA $P = 4s$ COPY SUBSTITUTE ANSWER
3 Evaluate. $2^3 + 3^2$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Write 12:30 am in 24-hour time.	FORMULA 24-hour time: pm hours add 12 (1:00 pm = 13:00); 12:00 am = 00:00 COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = x + 1$. Write the coordinates of B.  	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

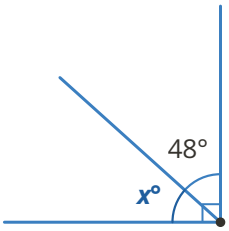
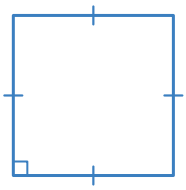
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the area of this square.  	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. $\frac{2}{5}$	FORMULA Fraction to decimal: numerator ÷ denominator COPY SUBSTITUTE ANSWER
4 Expand. $5(x + 2)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the range of these heights. cm: 165, 152, 171, 158, 160	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

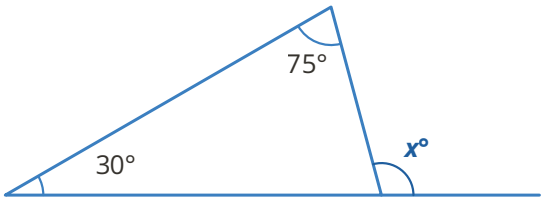
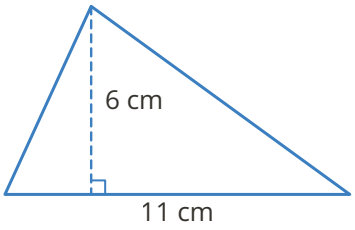
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 16 out of 64	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Simplify. 14 : 21	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 13, 19, 16, 10, 17	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

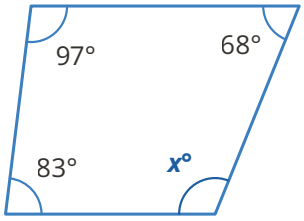
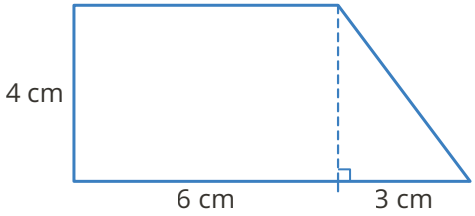
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{2}{5}$ of 45 kg	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Share 48 marbles in the ratio 3 : 5. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the median of these scores. Scores: 6, 2, 9, 4, 10, 8	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

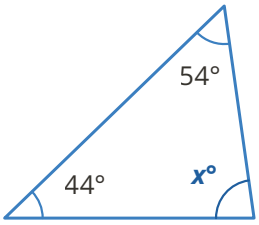
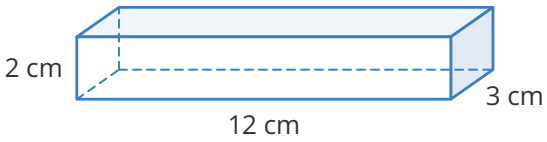
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism.  	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Calculate. 35% of 200 m	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Solve. $y - 15 = 8$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A spinner was spun 50 times. Find the experimental probability of blue. Colour: Red 12, Blue 18, Green 20	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

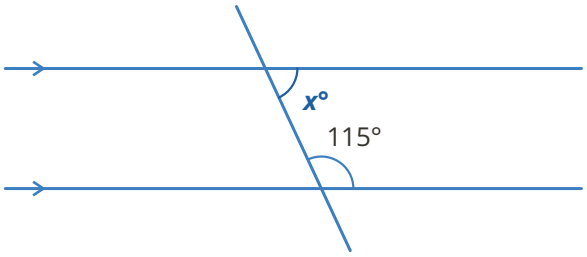
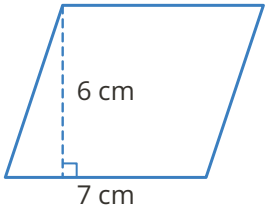
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 0.25×1000	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $a = 6$ and $b = 3$. $ab - 4$	FORMULA $ab - 4$ COPY SUBSTITUTE ANSWER
5 Find the probability that a try is not scored. $P(\text{try}) = \frac{2}{9}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

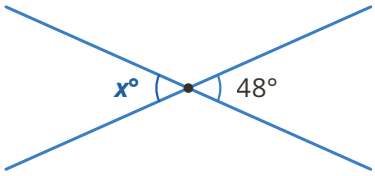
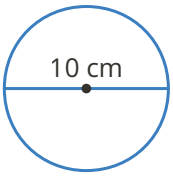
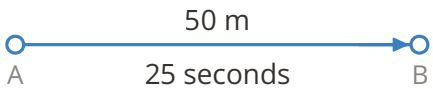
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. 64%	FORMULA Percentage to decimal: percentage ÷ 100 COPY SUBSTITUTE ANSWER
4 A swimmer swims from A to B. Find the average speed. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A letter is chosen at random from this word. Find $P(\text{M})$. MATHEMATICS	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

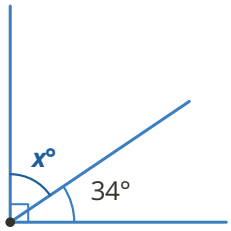
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Convert 2.5 L to millilitres.	FORMULA $1\ cm^3 = 1\ mL,\ 1\ L = 1000\ mL$ COPY SUBSTITUTE ANSWER
3 Evaluate. $40 \div 8 - 2 \times 9$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Expand. $2(x + 7)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 10 - 2x$. x: 1, 2, 3, 4 y: 8, ____, 4, ____	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

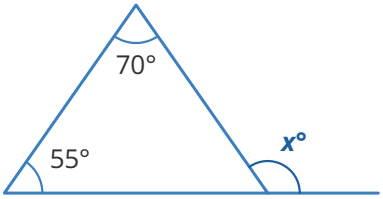
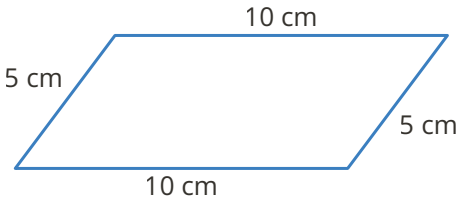
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the perimeter of this parallelogram.  	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Write as a percentage. 7 out of 35	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 A car travels 360 km on 30 L of fuel. Find the rate in kilometres per litre.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 Find the mean of these try tallies. Tries: 2, 0, 4, 1, 3	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

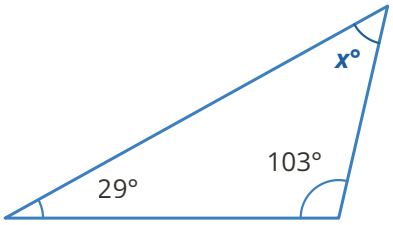
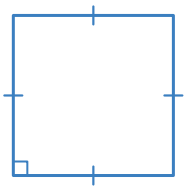
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this square.  	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{2}{3}$ of 27	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Surf club training starts at 06:30 and finishes at 07:05. How long does it last?	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 Find the range of these scores. Scores: 9, 3, 12, 7, 5	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

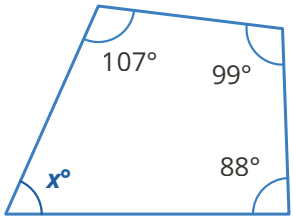
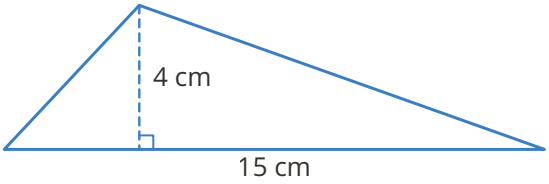
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $5^2 + 2^4$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Solve. $6y + 4 = 40$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the mode of these scores. 5 8 3 8 6 2 8	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

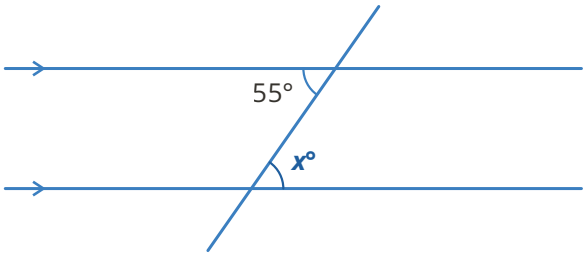
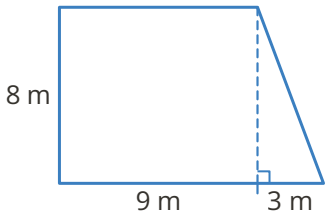
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Alternate angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 80% of \$45	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $h = -1$ and $g = 9$. $4h + 2g$	FORMULA $4h + 2g$ COPY SUBSTITUTE ANSWER
5 A six came up 15 times in 60 rolls of a die. Find the experimental probability of rolling a six.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

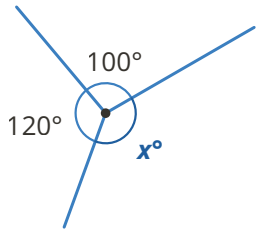
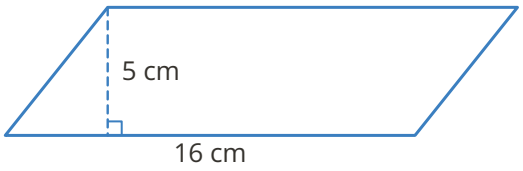
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 0.6	FORMULA Decimal to fraction: write over 10, 100 or 1000, then simplify COPY SUBSTITUTE ANSWER
4 Simplify. 45 : 60	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is white. Colour: Red, White Marbles: 5, 3	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER